Module 4: Lesson 4

Multivariate RNNs

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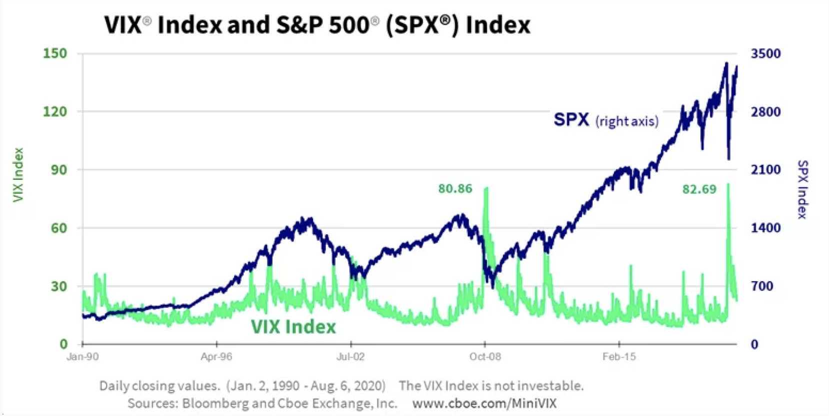
Outline

- The S&P500 and the VIX
- Multivariate RNNs

The S&P500 and the VIX

For our next prediction problem, we will use a multivariate RNN. What does this mean? Simply that it contains more than one input feature. This makes a lot of sense, especially if we can get our hands on 'leading' data:



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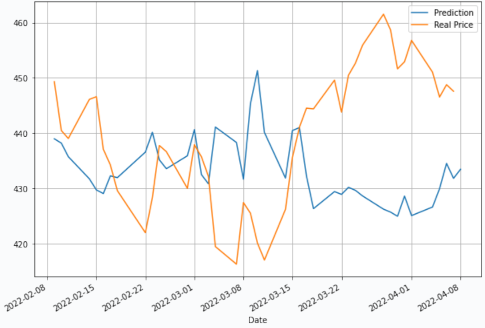
The S&P500 and the VIX

In our multivariate RNN, we will aim to predict the next day's price of the SPY (ETF over the S&P500 index) using past prices of SPY and prices (or levels) of the VIX.

Sadly, our results are not so good...



Note that the training of this network requires some random initialization so the graph may look different!

Summary of Lesson 4

In Lesson 4, we:

- Explored the relationship between a market index such as S&P500 and its volatility index (VIX)
- Used both inputs in an RNN
- Worked with the multivariate case of RNNs

TO DO NEXT: Go to the accompanying Python notebook to implement your first multivariate RNN!

This is all for Module 4. In the next module, we will deal with two 'modern' types of RNNs: gated recurrent units (GRUs) and long short-term memory (LSTM). See you there!